CISC 2200 Fall 2026
Lab 1 70 points **Due Time: 9/29/2026**

Programming Requirements:

Your assignment is to extend an UnsortedType Abstract Data Type, which is implemented with an array. You are required to complete the design of Student class (**Student.h, Student.cpp**), ItemType class (**ItemType.h**), and extend the implementation of UnsortedListByArray class (**UnsortedListByArray.h, UnsortedListByArray.cpp),** and you need to write a test driver **Lab1.cpp** to test your extended ADT.

In this project, please compare two strings as C++ builtin function does. Keep all functions we have discussed in class, additional functions should be added.

1) Implementation of Student class. Student.h is given, complete the four setter and four getter functions in Student.cpp. (**16 point**s)
2) Complete the function ComparedTo( ) in ItemType.h, which compares two Fordham students by their FIDN. (**5 points**)
3) Add a new member function to insert an item into the list. You are not allowed to call existing member functions of the UnsortedListByArray Class to complete this task. Both UnsortedListByArray.h and UnsortedListByArray.cpp should be updated. (**12 points**)

   The member function **InsertItem ( )** we discussed in class does not check whether the new item exists in the list. If the user wants to maintain a list with unique items, checking should be done before calling the insert function. Let's add a new insert function which checks the duplicate copy for the user before inserting the item. The following is the description of the new function.

bool NewInsertItem (ItemType newItem)

Function	Insert the item into the list if the list does not have it. The function will return True if the item is inserted successfully; False if the list has the same item.
Precondition	List has been initialized.
Postcondition	List should have just one copy of the new item.

Example (Only FIDNs are shown):

Case A: Before inserting Student with FIDN **"A001"**, UnsortedListByArray list L :

A005	A110	A002	**A001**	A111

After inserting, UnsortedListByArray list L

A005	A110	A002	**A001**	A111

Case B: Before inserting **"Student: Tom(A001)/CS"**, UnsortedListByArray list L :

A005	A110	A002

After inserting **"A001"**, UnsortedListByArray list L:

A005	A110	A002	**A001**

4) Add a new member function to split the unsorted list into two sublists, each of them is an unsorted list. Both UnsortedListByArray.h and UnsortedListByArray.cpp should be updated. (**12 points**)

void SplitList(ItemType item,
UnsortedListByArray & listOne, UnsortedListByArray & listTwo)

Function	Divides list into two lists according to the key of item
Precondition	List has been initialized and is not empty. listOne and listTwo are initialized but empty.
Postcondition	listOne contains all the items of list whose keys are **less than or equal** to item's key; listTwo contains all the items of the list whose keys are greater than the item's key.

Example: Before split, UnsortedListByArray list L :

A005	A110	A002	A001	A112	A008	A111

If ItemType item's key is "**A009**", after split we got

UnsortedListByArray listOne

A005	A002	A001	A008

UnsortedListByArray listTwo

A110	A112	A111

5) A driver file **Lab1.cpp** should be completed. The file will process inputs of students' information from the keyboard, and test new added functions of UnsortedListByArray class. TestUnsortedList.cpp should be used as the template of this driver file. Test cases are shown in **Out.txt.pdf**. (**25 points**)

* **FileZilla** app is recommended to do the file transfer (https://filezilla-project.org/)

Submit your files on BB: **UnsortedListByArray.h, UnsortedListByArray.cpp, ItemType.h, Student.h, Student.cpp, and Lab1.cpp.**